

Mathematics Bootcamp

Optimization: Finding the Best Feasible Choice

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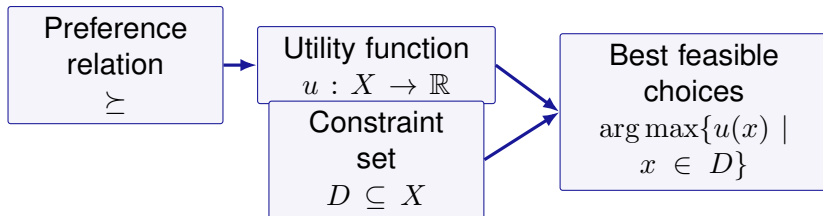
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Today

- What an optimization problem asks
- Why first-order conditions are necessary
- Equality constraints and Lagrange multipliers
- Inequality constraints and Kuhn–Tucker conditions
- Consumer demand: smooth optima, corners, and kinks

We finish the mathematics before applying it to consumer choice.

Optimization Completes the Path from Preferences to Choice



The Question

Which feasible choices are at least as good as every other feasible choice?

An increasing transformation of u changes utility numbers but not the set of maximizers.

The Maximum and the Maximizers Are Different Objects

Consider

$$\max\{f(x) \mid x \in D\}.$$

Maximum Value

If a solution exists, then

$$\max\{f(x) \mid x \in D\} = f(x^*) \in \mathbb{R}.$$

This is a number.

Maximizers

$$\begin{aligned} & \arg \max\{f(x) \mid x \in D\} \\ &= \{x \in D : f(x) \geq f(y) \ \forall y \in D\}. \end{aligned}$$

This is a set of choices.

The argmax may be empty, contain one point, or contain many points.

Existence Comes Before Calculation

Open feasible set

$$\max_{x \in (0,1)} x$$

Supremum 1, but no maximizer.

Unbounded objective

$$\max_{x \geq 0} x$$

No finite maximum.

Discontinuity

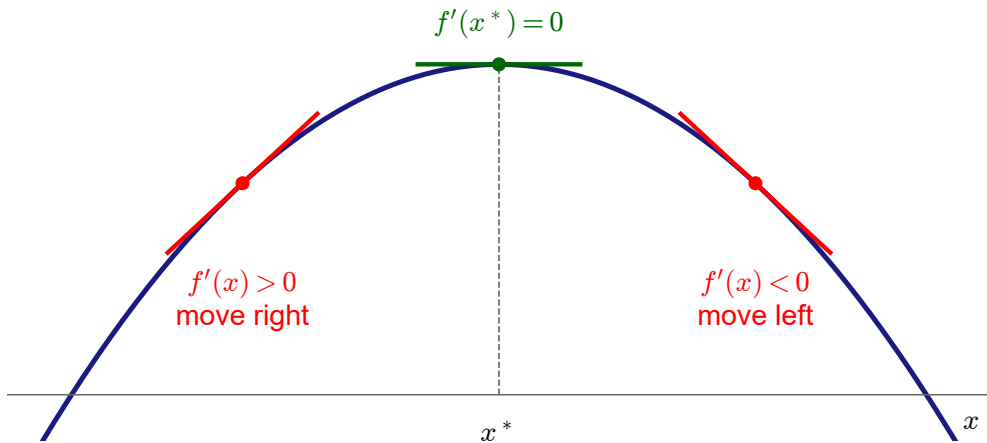
$$f(x) = \begin{cases} x, & x < \frac{1}{2}, \\ x - 1, & x \geq \frac{1}{2}. \end{cases}$$

On $[0, 1]$, the supremum is not attained.

Extreme Value Theorem

If $D \subseteq \mathbb{R}^n$ is nonempty and compact and $f : D \rightarrow \mathbb{R}$ is continuous, then f attains a maximum and a minimum on D .

If the Slope Is Nonzero, Improvement Is Possible



Why the First-Order Condition Is Necessary

First-Order Necessary Condition

If $x^* \in \text{int } D \subseteq \mathbb{R}$ is a local maximizer of f on D and f is differentiable at x^* , then $f'(x^*) = 0$.

(1) For sufficiently small $h > 0$, local optimality gives

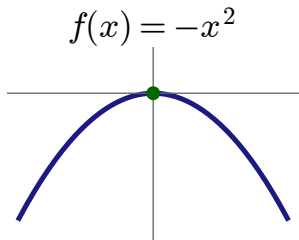
$$f(x^* + h) \leq f(x^*) \implies \frac{f(x^* + h) - f(x^*)}{h} \leq 0.$$

(2) For sufficiently small $h < 0$, the numerator is still nonpositive but the denominator is negative, so

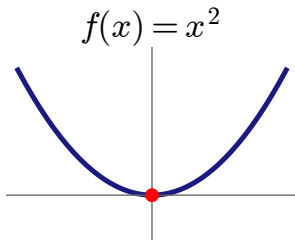
$$\frac{f(x^* + h) - f(x^*)}{h} \geq 0.$$

(3) The two one-sided limits are equal because f is differentiable. Hence $f'(x^*) \leq 0$ and $f'(x^*) \geq 0$, so $f'(x^*) = 0$.

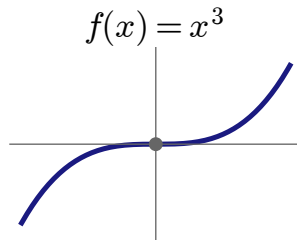
A Critical Point Is Only a Candidate



local and global
maximum



local and global
minimum



neither

All three satisfy $f'(0) = 0$. Moreover, a boundary maximizer need not satisfy $f'(x) = 0$ because movement may be possible in only one direction.

The Gradient Tests Every Direction at Once

For a direction $d \in \mathbb{R}^n$,

$$f(x + td) - f(x) \approx t \nabla f(x) \cdot d.$$

If x^* is an interior local maximum, define

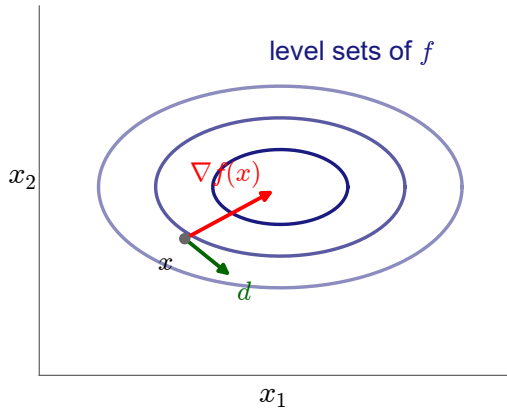
$$\phi_d(t) = f(x^* + td).$$

Then $t = 0$ locally maximizes ϕ_d , so

$$\phi'_d(0) = \nabla f(x^*) \cdot d = 0 \quad \text{for every } d.$$

Therefore,

$$\boxed{\nabla f(x^*) = 0.}$$



Curvature Turns Candidates into Conclusions

Local Conclusion

Suppose f is twice differentiable and

$$\nabla f(x^*) = 0.$$

If $\nabla^2 f(x^*)$ is negative definite, then x^* is a strict local maximizer.

Negative semidefiniteness at x^* is necessary for a smooth local maximum, but may not be sufficient.

First-order conditions generate candidates; curvature and concavity tell us what those candidates mean.

Global Conclusion

Suppose D is convex and f is differentiable and concave on D . Then

$$\nabla f(x^*) = 0 \implies x^* \in \arg \max \{f(x) \mid x \in D\}.$$

If f is strictly concave, the maximizer is unique whenever it exists.

A Constraint Removes Directions

Consider

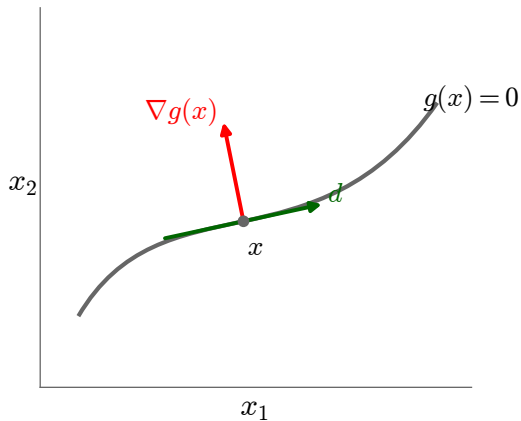
$$\max_x f(x) \quad \text{subject to} \quad g(x) = 0.$$

A direction d tangent to the constraint satisfies, to first order,

$$\nabla g(x) \cdot d = 0.$$

At a constrained optimum, the objective must not increase along any such feasible direction.

The gradient $\nabla g(x)$ is normal to the constraint curve.



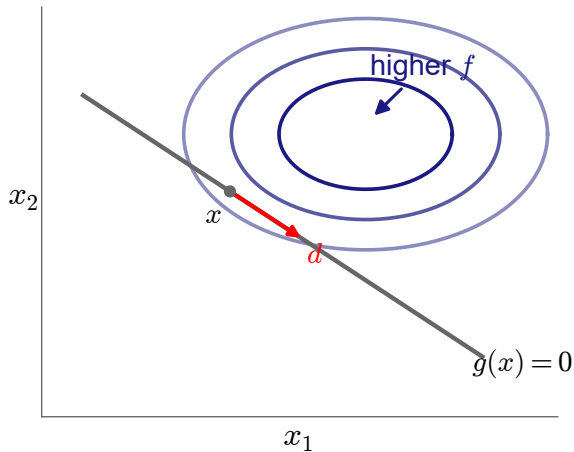
If the Curves Cross, Improvement Remains Possible

If the feasible curve crosses an objective level curve, one of the two feasible directions leads toward a higher level.

At the marked point,

$$\nabla f(x) \cdot d > 0.$$

A small feasible movement in direction d raises the objective, so the point cannot be optimal.



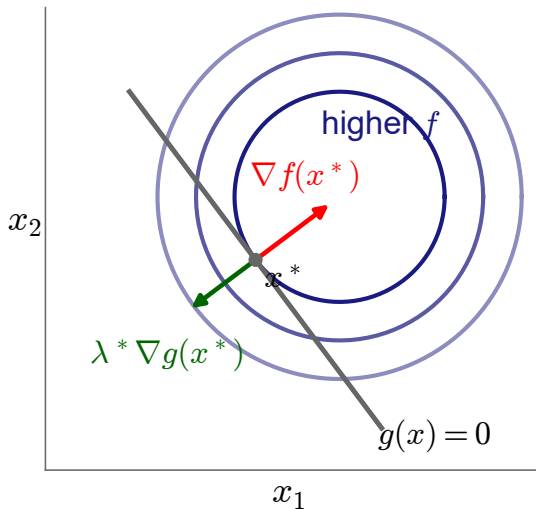
At a Smooth Constrained Optimum, the Normals Align

At tangency, the objective level curve and the feasible curve share the same tangent direction. Their normal vectors must therefore be parallel:

$$\nabla f(x^*) + \lambda^* \nabla g(x^*) = 0.$$

Equivalently, every feasible tangent direction d satisfies

$$\nabla f(x^*) \cdot d = 0.$$



The Constraint Cuts a Feasible Curve from the Surface

Consider

$$f(x) = 4 - [(x_1 - 1)^2 + (x_2 - 1)^2],$$

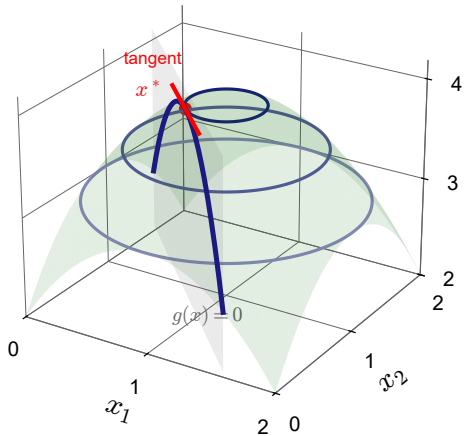
$$g(x) = x_1 + x_2 - 1.6 = 0.$$

The gray plane cuts a blue feasible curve from the green objective surface. Maximizing f on $g(x) = 0$ means choosing its highest point:

$$x^* = (0.8, 0.8).$$

At x^* the red tangent is horizontal:

$$d = (1, -1), \quad \nabla f(x^*) \cdot d = 0.$$



The Lagrangean Packages Tangency and Feasibility

For

$$\max_x f(x) \quad \text{subject to} \quad g(x) = 0,$$

define

$$L(x; \lambda) = f(x) + \lambda g(x).$$

Stationarity

$$\begin{aligned}\nabla_x L(x^*; \lambda^*) &= 0, \\ \nabla f(x^*) + \lambda^* \nabla g(x^*) &= 0.\end{aligned}$$

Feasibility

$$g(x^*) = 0.$$

The candidate must satisfy the constraint.

These equations generate candidates. Concavity or direct comparison is still needed to establish global optimality.

Regular Constraints Make the Multiplier Well Defined

Theorem of Lagrange

Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and let U be open. If x^* is a local optimum of f on $D = U \cap \{x : g(x) = 0\}$ and $\nabla g(x^*) \neq 0$, then there exists a unique $\lambda^* \in \mathbb{R}$ such that

$$\nabla f(x^*) + \lambda^* \nabla g(x^*) = 0.$$

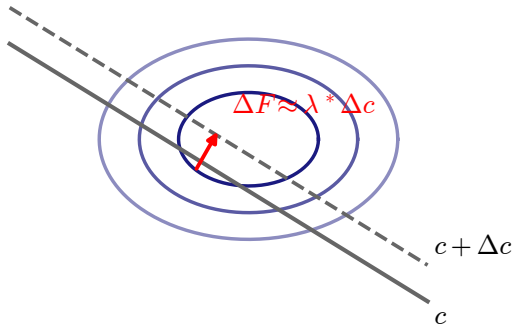
For $g = (g_1, \dots, g_k)$, let J_g denote its Jacobian. The constraint qualification is $\rho(J_g(x^*)) = k$.

- Regularity guarantees existence of the multiplier at a local optimum.
- Independence makes its coefficients unique at that fixed x^* .
- It does not guarantee that the optimizer itself is unique.

A Multiplier Measures the Value of Relaxing a Constraint

Let $g(x; c) = g(x) + c$, let $x^*(c)$ solve $\max\{f(x) \mid g(x; c) = 0\}$, and define $F(c) = f(x^*(c))$. If $x^*(c)$ varies smoothly, then $F'(c) = \lambda^*(c)$.

- λ^* is first a coefficient aligning two normal vectors.
- With the additional smoothness, it is the marginal optimized value of relaxing the constraint.
- Its value depends on the chosen representation of the objective.



Inequality Constraints Are Effective or Slack

Write each inequality as $h_i(x) \geq 0$. At a candidate x^* :

Slack Constraint

$$h_i(x^*) > 0.$$

The constraint is slack. Small changes in its boundary do not affect the local choice, so $\lambda_i^* = 0$.

Effective Constraint

$$h_i(x^*) = 0.$$

The boundary restricts feasible directions and may carry a positive multiplier.

Both cases are summarized by

$$\lambda_i^* \geq 0, \quad h_i(x^*) \geq 0, \quad \lambda_i^* h_i(x^*) = 0.$$

The Kuhn–Tucker Conditions Are One-Sided

Theorem of Kuhn and Tucker

Let $f, h_1, \dots, h_\ell$ be continuously differentiable, let U be open, and define $D = U \cap \{x \in \mathbb{R}^n : h_i(x) \geq 0, i = 1, \dots, \ell\}$. Suppose x^* is a local maximum of f on D . Let $E = \{i : h_i(x^*) = 0\}$ and suppose $\rho(J_{h_E}(x^*)) = |E|$. Then there exists $\lambda^* \in \mathbb{R}^\ell$ such that

$$[\text{KT-1}] \quad \lambda_i^* \geq 0, \quad \lambda_i^* h_i(x^*) = 0, \quad i = 1, \dots, \ell,$$

$$[\text{KT-2}] \quad \nabla f(x^*) + \sum_{i=1}^{\ell} \lambda_i^* \nabla h_i(x^*) = 0.$$

Only effective normals contribute; they generate a nonnegative cone, not a linear span.

Consumer Choice Is the Same Optimization Problem

A consumer chooses $x \in \mathbb{R}_+^n$, has income $I > 0$, and faces prices $p \in \mathbb{R}_{++}^n$.

Walrasian Budget Set

$$B(p, I) = \{x \in \mathbb{R}_+^n : p \cdot x \leq I\}.$$

Walrasian Demand Correspondence

$$x(p, I) = \arg \max \{u(x) \mid x \in B(p, I)\}.$$

A selected optimum is written $x^* \in x(p, I)$.

Demand is a correspondence because several bundles may maximize utility.

The Assumptions from Day 3 Now Do Specific Jobs

Assumption	Consequence for consumer choice
$p \gg 0$ and $I > 0$	$B(p, I)$ is nonempty, compact, and convex.
Continuity of u	$x(p, I)$ is nonempty and compact.
Strict quasiconcavity of u	At most one bundle maximizes u on the convex budget set.
Local nonsatiation	Every optimal bundle exhausts the budget.

These are distinct conclusions: existence of a maximizer, uniqueness of the maximizer, and existence of multipliers are different questions.

Local Nonsatiation Makes the Budget Bind

Walras' Law for Individual Demand

If preferences are locally nonsatiated and $x^* \in x(p, I)$, then

$$p \cdot x^* = I.$$

- (1) Suppose instead that $p \cdot x^* < I$. There is positive spending slack.
- (2) Because prices are finite, every sufficiently close bundle remains affordable.
- (3) Local nonsatiation gives a sufficiently close bundle y with $y \succ x^*$.
- (4) Then $y \in B(p, I)$ and $y \succ x^*$, contradicting $x^* \in x(p, I)$.

Hence the inequality constraint may be treated as the equality $p \cdot x = I$ when solving a locally nonsatiated consumer problem.

A Money-Neutral Reallocation Is a Feasible Direction

Move $\delta > 0$ units of money from good j to good i :

$$\Delta x_i = \frac{\delta}{p_i}, \quad \Delta x_j = -\frac{\delta}{p_j}.$$

The change in expenditure is zero:

$$p_i \Delta x_i + p_j \Delta x_j = \delta - \delta = 0.$$

To first order, the utility change is

$$\Delta u \approx \delta \left(\frac{\partial u(x)/\partial x_i}{p_i} - \frac{\partial u(x)/\partial x_j}{p_j} \right).$$

If the expression in parentheses is positive, the current bundle is not optimal.

At an Interior Optimum, No Money Switch Pays

If both goods are consumed in strictly positive amounts, both reallocation directions are feasible. Therefore,

$$\frac{\partial u(x^*)/\partial x_i}{p_i} = \frac{\partial u(x^*)/\partial x_j}{p_j} = \lambda^*.$$

Equivalently,

$$\nabla u(x^*) = \lambda^* p, \quad \frac{\partial u(x^*)/\partial x_i}{\partial u(x^*)/\partial x_j} = \frac{p_i}{p_j}.$$

Interpretation

Each consumed good produces the same marginal utility per unit of money. Under the smooth value-function assumptions, $\lambda^* = \partial v(p, I)/\partial I$.

A Monotone Transformation Simplifies Cobb–Douglas

Let

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, \quad 0 < \alpha < 1.$$

Because $\log$ is strictly increasing,

$$\arg \max \{u(x) \mid x \in B(p, I)\} = \arg \max \{\alpha \log x_1 + (1 - \alpha) \log x_2 \mid x \in B(p, I)\}.$$

With local nonsatiation, use the binding budget and write

$$L(x; \lambda) = \alpha \log x_1 + (1 - \alpha) \log x_2 + \lambda(I - p_1 x_1 - p_2 x_2).$$

The maximizing bundle is unchanged by the transformation; the numerical value of the multiplier is not.

Cobb–Douglas Demand Allocates Fixed Shares of Income

The first-order conditions are

$$\frac{\alpha}{x_1} = \lambda^* p_1, \quad \frac{1 - \alpha}{x_2} = \lambda^* p_2, \quad p_1 x_1 + p_2 x_2 = I.$$

Multiplying the first two equations by x_1 and x_2 gives

$$\alpha = \lambda^* p_1 x_1, \quad 1 - \alpha = \lambda^* p_2 x_2.$$

Summing and using the budget yields $\lambda^* = 1/I$, so

$$x_1(p, I) = \frac{\alpha I}{p_1}, \quad x_2(p, I) = \frac{(1 - \alpha) I}{p_2}.$$

The consumer spends the shares α and $1 - \alpha$ of income on the two goods.

Perfect Substitutes Turn Tangency into a Comparison

Let $u(x_1, x_2) = ax_1 + bx_2$, where $a, b > 0$. Marginal utility per unit of money is constant:

$$\frac{a}{p_1} \quad \text{and} \quad \frac{b}{p_2}.$$

Walrasian Demand

$$x(p, I) = \begin{cases} \{(I/p_1, 0)\}, & a/p_1 > b/p_2, \\ \{(0, I/p_2)\}, & a/p_1 < b/p_2, \\ \{x \in \mathbb{R}_+^2 : p \cdot x = I\}, & a/p_1 = b/p_2. \end{cases}$$

Unless the two returns are equal, the optimum is a corner rather than a tangency.

Leontief Utility Is Solved at the Kink

Let

$$u(x_1, x_2) = \min\{ax_1, bx_2\}, \quad a, b > 0.$$

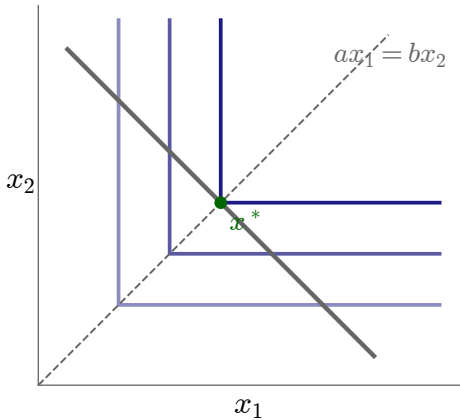
Unmatched units do not raise utility, so an optimum satisfies

$$ax_1 = bx_2.$$

Together with $p_1x_1 + p_2x_2 = I$,

$$x_1(p, I) = \frac{bI}{bp_1 + ap_2}, \quad x_2(p, I) = \frac{aI}{bp_1 + ap_2}.$$

Ordinary smooth tangency conditions do not apply at the economically relevant kink.



Kuhn–Tucker for the Consumer Problem

Write the constraints as $h_i(x) = x_i$ for $i = 1, \dots, n$ and $h_{n+1}(x) = I - p \cdot x$. With nonnegative multipliers,

$$L(x; \lambda) = u(x) + \sum_{i=1}^n \lambda_i x_i + \lambda_{n+1}(I - p \cdot x).$$

With local nonsatiation, the budget binds. For every good, stationarity and complementary slackness give

$$\frac{\partial u(x^*)}{\partial x_i} + \lambda_i^* - \lambda_{n+1}^* p_i = 0, \quad \lambda_i^* x_i^* = 0.$$

If $x_i^* > 0$, then $\lambda_i^* = 0$ and

$$\frac{\partial u(x^*)/\partial x_i}{p_i} = \lambda_{n+1}^*.$$

If $x_i^* = 0$, then

$$\frac{\partial u(x^*)/\partial x_i}{p_i} \leq \lambda_{n+1}^*.$$

Every Case Asks the Same Question

Central Question

Is there a feasible direction that raises the objective?

1. Write the objective and the feasible set.
2. Establish existence and uniqueness separately.
3. Identify whether the candidate is interior, on a smooth boundary, at a corner, or at a kink.
4. Generate candidates using first-order conditions, the Lagrangean method, Kuhn–Tucker, or geometry.
5. Verify feasibility, complementary slackness, and global optimality.

First-order conditions reduce the search; assumptions and verification finish it.